

Report of attendance

ATRIUM Training School

Róbert Szederkényi

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ELTE Research Centre for the
Humanities, Institute of
Archaeology

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I currently work as a researcher-archivist at the Institute of Archaeology, ELTE Research Centre for the Humanities, in Budapest. My work combines archival research, data curation, research support, artefact photography, and the management of large analogue and digital archaeological collections.

Between 14 and 18 September 2026, I attended the ATRIUM Training School “Computer Vision in Archaeology” in Brno, Czech Republic. The training school formed part of the trans-national access scheme provided by the ATRIUM Project and was hosted by the Archaeological Information System of the Czech Republic (AIS CR) research infrastructure at the Institute of Archaeology of the Czech Academy of Sciences, Brno. The school was co-organised by the MAIA COST Action, which also provided funding for several of the instructors.

From the beginning, the training school felt very well organised and welcoming. The week included lectures, practical exercise and plenty of informal discussion, and the pace never felt rushed. I appreciated the friendly attitude of the organisers and instructors, who were always available when someone needed help or further explanation. The facilities and common areas were excellent, and the regular breaks gave participants enough time to discuss the sessions. The social activities also helped create a relaxed atmosphere and made it easier to connect with both the teaching team and the other participants.

Monday

The first day provided an introduction to computer vision and its possible applications in archaeology. We discussed different computer-vision tasks and worked with CLIP for zero-shot classification of archaeological photographs. Working with CLIP made it quite clear how much the result can depend on the labels and wording given to the model, and also showed where the limits of a pre-trained model begin to appear. The programme also introduced the Python fundamentals required for image processing, including NumPy, Matplotlib and OpenCV, which are widely used libraries for numerical data processing, visualisation and computer-vision tasks. A practical exercise used an archaeological photograph to separate an object from its background and perform basic measurements.

Another session focused on coding with AI agents. Here the emphasis was on giving clear instructions, documenting the process and creating procedures that could later be reused. From an archaeological point of view, I found this especially relevant for work involving large image collections. Classification, measurement, annotation and quality control often contain repetitive steps that can potentially be supported or partly automated. At the same time, the exercises made clear that the output of such methods is usually only another stage of the research process. A model may group images, suggest classifications or highlight patterns, but these results still have to be examined and interpreted archaeologically.

Tuesday

On the second day, the focus shifted to archaeological artefact photographs, datasets and annotation. Annotation means adding structured information to images so that they can later be used for training and evaluating computer-vision models. Using CVAT, an online tool designed for image and video annotation, we marked archaeological objects in photographs, assigned them to predefined categories and recorded additional characteristics such as fragmentation or damage. In this way, an ordinary photograph can be transformed into structured training data by telling the model not only what is present in the image, but also where the relevant object is and what characteristics it has.

We also became familiar with annotation formats such as COCO, YOLO and Pascal VOC. These are different ways of storing the labels, object locations and other information associated with the images so that they can be used by different computer vision tools and machine learning workflows. The session also demonstrated Meta's SAM 2, which can assist image segmentation by suggesting the boundaries of objects. This can considerably reduce the amount of manual work required, although the researcher still needs to check and correct the resulting annotations. One point that became clear during the exercises was how strongly the later result depends on the quality of the annotation. Inconsistent labels or poorly defined object boundaries do not simply create untidy data, they directly affect what the model can learn. Creating a well-annotated dataset therefore requires considerable manual work at the beginning, but this work does not necessarily have to be repeated for every later analysis. The same dataset may later be useful for classification, measurements, shape analysis, filtering or questions that were not even part of the original project.

The exercises showed several problems that are relevant to archaeological datasets. These included class imbalance, inconsistencies between different vocabularies and classification systems, and the large amount of potentially useful metadata that is often missing because recording it manually would require considerable manual labour. The exercise therefore demonstrated that the quality, consistency and structure of human annotation have a direct influence on what can later be achieved with machine learning models. We also worked with a larger archaeological training dataset in COCO format and looked at how the images and their annotations were organised for training and evaluation. For me, this was one of the practical arguments for investing time in annotation: once an image collection has been consistently segmented and described, later analyses do not necessarily have to start again from the original photographs.

In the afternoon, participants had the opportunity to present their own research topics in which they had used AI-related methods. These demonstrated the considerable variety of ways in which computer vision and machine learning are already being explored in archaeology and related fields. These presentations were particularly useful because they showed that there is no single "archaeological AI workflow". Similar families of methods can be adapted to very different materials and research questions.

Wednesday

The third day focused on computer vision and machine learning workflows for microscopic archaeological images, using experimental lithic use-wear data as a case study. The images showed polish produced on flint tools through contact with different materials such as wood and bone. Before any machine-learning analysis, however, we first had to consider whether the images were actually comparable. Differences in image dimensions, illumination, contrast, focus and other acquisition conditions can strongly influence the following computer analysis.

We worked through several stages of image preprocessing and standardisation. We first examined the properties and pixel-intensity distributions of the images and then experimented with resizing, histogram matching and histogram equalisation to make images acquired under different conditions more comparable. Using edge information and segmentation techniques, we attempted to identify the areas containing the most useful surface detail and separate them from blurred or irrelevant regions. We then used thresholding method to distinguish brighter polished areas from the surrounding surface and calculate their proportion within an image.

The exercises also covered image noise and data augmentation. We tested how filtering can remove artificial image noise without destroying relevant surface information, and discussed how transformations such as rotation or rescaling can increase the diversity of small training datasets. At the same time, the exercises emphasised that such transformations must remain archaeologically meaningful: not every technically possible modification is appropriate for every dataset.

Several parts of the practical work contained incomplete or deliberately problematic code that we had to complete, test and correct ourselves with the help of documentation and AI-assisted coding. This was useful not only for learning individual Python functions, but also for understanding why each processing step was being applied and how changes in parameters affected the archaeological information retained in the image. The overall workflow demonstrated that successful machine learning analysis begins well before model training: the quality, comparability and relevance of the input data can be just as important as the choice of algorithm itself.

Thursday

Thursday introduced two different applications of computer vision in archaeology. The morning session focused on archaeological remote sensing using satellite data. This was completely new to me and therefore especially interesting. We worked with Sentinel-2 multispectral imagery and explored how information beyond ordinary RGB photography can be used to examine differences in vegetation and surface conditions. Buried archaeological features may influence soil moisture and plant growth, and these differences can sometimes appear as cropmarks in satellite imagery.

Using the Bronze Age fortified site of Cornești-Iarcuri as a case study, with its enormous extent makes it particularly suitable for remote-sensing approaches, we worked with spectral bands, vegetation indices such as the Normalized Difference Vegetation Index (NDVI), and methods for enhancing local vegetation anomalies and line-like features. An important point was that the same archaeological feature does not necessarily produce the same spectral response across different parts of the landscape. Depending on local conditions, vegetation above a buried feature may appear either stronger or weaker than its surroundings, so the analysis had to take local context into account rather than relying on a single threshold or visual characteristic.

These different sources of information were then combined using a supervised Random Forest classifier. The model was trained with examples identified by researchers and used spectral information, vegetation response, local contrast and spatial shape together rather than relying on a single image property. The first classification produced a number of false positives, including field boundaries, agricultural tracks and other vegetation patterns. These errors were then used as additional training examples in a process known as hard-negative mining, demonstrating how the model could be improved iteratively.

The resulting probability map was converted into GIS-compatible polygons representing possible archaeological features. Importantly, these were not interpreted as archaeological features simply because the model had detected them. They remained candidates to be checked. In this sense, the computer vision stage reduced the amount of landscape that needed detailed inspection without replacing the archaeological interpretation itself.

The afternoon session focused on rock-art imagery and the challenges of detecting small archaeological features in very large images. We examined how object-detection models can lose small details when high-resolution photographs are reduced in size, and how tiled inference can address this problem by analysing the image in smaller sections and combining the predictions afterwards. We also looked at the different kinds of output produced by the model, including object locations, segmentation masks, class labels and confidence scores.

The session also showed how these results can be connected to broader digital documentation workflows. As an example of what such work can ultimately produce, we explored an interactive 3D rock-art model in the Swedish Rock Art Research Archives. Seeing the processed material in this form was particularly useful because it demonstrated that computer-vision outputs do not have to remain isolated detections or marked-up images: they can become part of a richer digital research environment in which archaeological features can be examined, compared and spatially contextualised.

As in several other sessions during the week, Mentimeter questions and short interactive exercises were used to involve participants directly in the discussion. This helped make the sessions less passive and provided opportunities to compare interpretations and methodological choices during the exercises.

Taken together, the two sessions provided a good example of how computer-vision methods can be integrated into existing archaeological workflows. In both cases, the model produced an intermediate result rather than a final archaeological conclusion, and these outputs still required checking, interpretation and contextualisation by the researcher.

Friday

The final day brought together several of the themes from the week through work with coin photographs and use of combined methods and vision-language models. One of the lessons learnt was that archaeological image analysis does not always require the latest or most specialised AI model. Older algorithms, as well as methods originally developed for different purposes, can still be highly effective when applied thoughtfully to a particular research problem. This encouraged a more flexible and problem-oriented approach rather than treating AI as a universal solution.

Overall, the training school provided me with a wide range of methods, software tools and good practices that I can directly apply in my own work. The examples demonstrated not only what computer vision and machine learning can achieve, but also the importance of well-structured datasets, consistent annotation, clear documentation and reproducible workflows. These principles are directly relevant to the digitised archaeological collections and research datasets that I manage, and I can see clear possibilities for applying them to image classification, metadata enrichment, quality control and improved searchability.

The discussions with the instructors and the AIS CR team were equally valuable. They provided an opportunity to exchange experiences about archaeological data management, digital infrastructures and the practical implementation of new technologies. I returned from the training with several useful software tools, methodological ideas and examples of good practice that I am confident I will be able to incorporate into my institutional work. I am very grateful for the opportunity to participate in the training school and for the support provided through the ATRIUM Transnational Access scheme.